

FO : Fully Observable

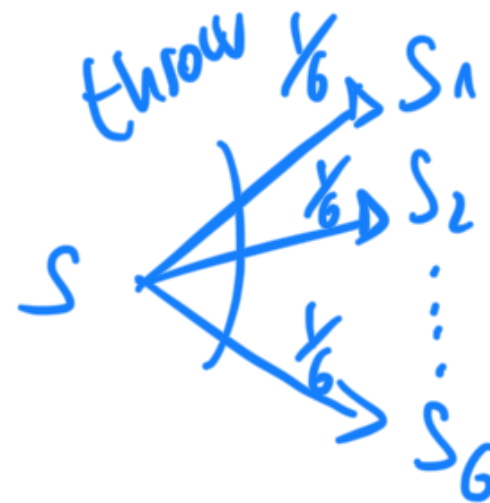
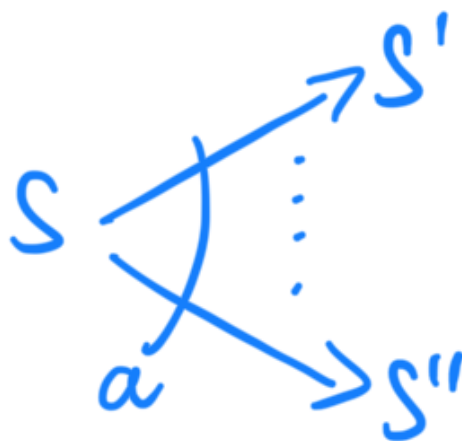
Ch. 17 AIMA

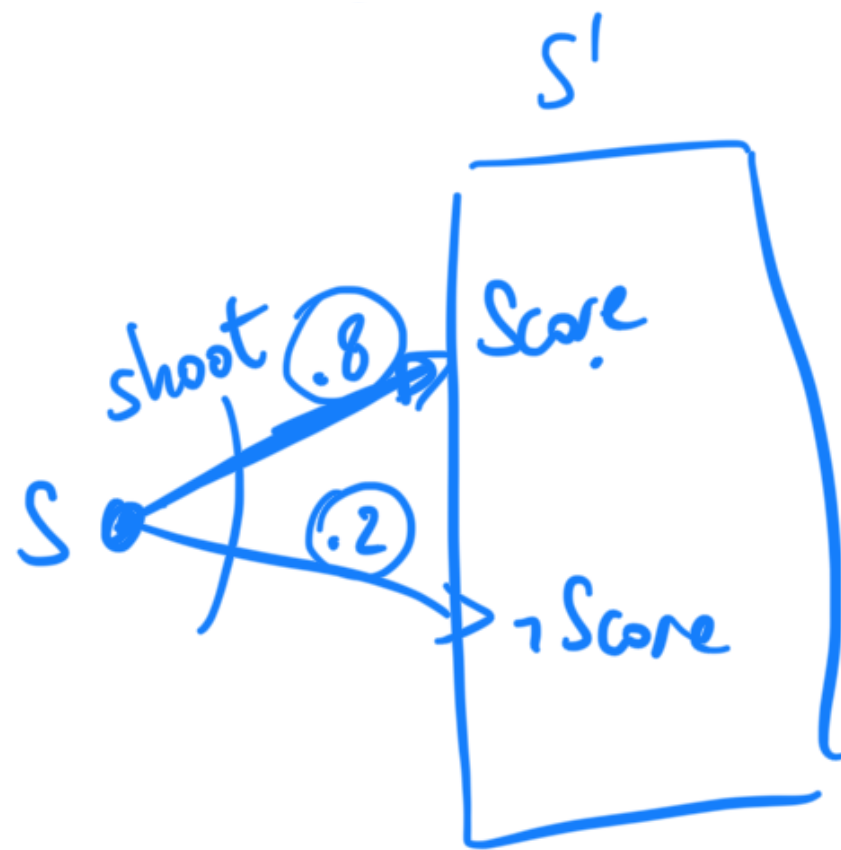
- ND



FOND Planning: qualitative
nondeterminism

- Stochastic





$$P(S' | S, a) = \begin{cases} 0.8, & \text{if } S = \text{Score} \\ 0.2 & \text{otherwise} \end{cases}$$

$$P(S' | S, a)$$

MARKOV DECISION PROCESS (MDP)

$$M = (S, A, \delta, R)$$

1) S : finite set of states •

2) . . .

2) A : finite set of actions •

3) δ : transition function: $\delta(s, a) = p(s' | s, a)$

4) R : reward function $R(s, a, s')$

$\uparrow$
 $s' \in S$

$$\delta(s, a) = \{s_1, \dots, s_6\}$$

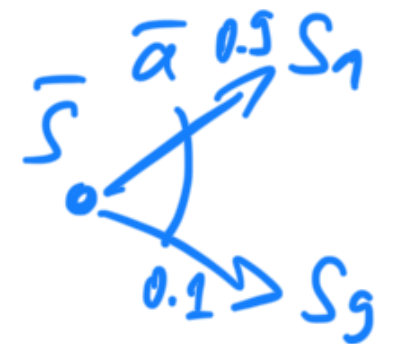
δ :

$$\delta(s, a) = p(\textcircled{s'} | \bar{s}, \bar{a}), \quad s' \in S$$

$$S = \{s_1, \dots, s_{100}\}$$

$$\delta(\bar{s}, \bar{a}) =$$

s'	prob
s_1	1



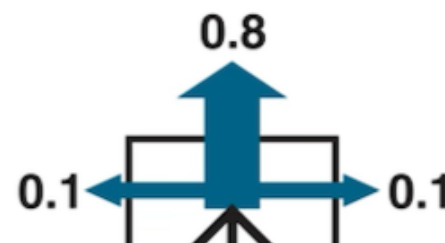
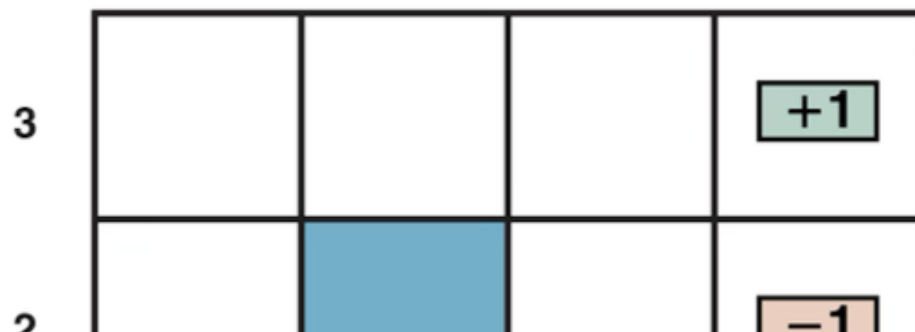
— | | |

$$\underline{\delta(s, a, s')}$$

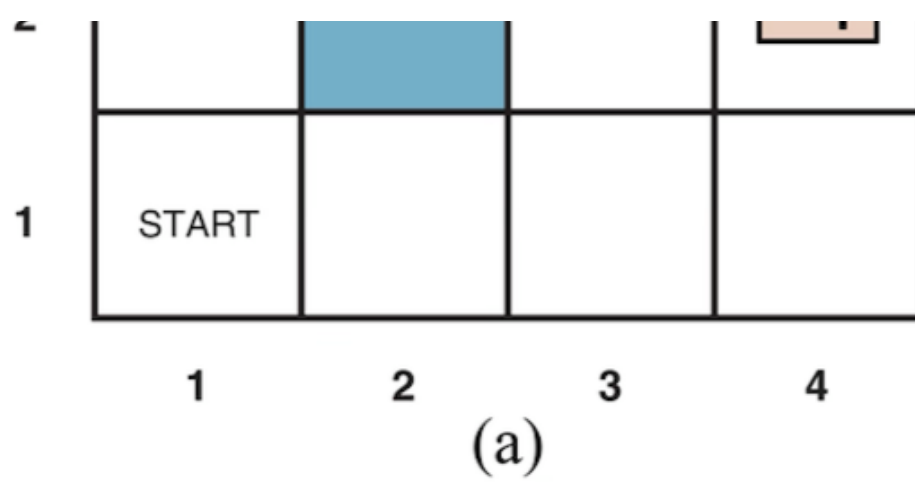
PLANNING with MDPs

- Stochastic / Probabilistic Planning
- Sequential Decision Making (improper)

$$\pi(s) = R$$



$$R(s, a, s') = \begin{cases} +1, & \text{if } s' = (4, 3) \\ -1, & \text{if } s' = (4, 2) \end{cases}$$



0, otherwise

$$\delta((1,1), UP) = \delta(s' \mid (1,1), UP) = \begin{cases} .8, & \text{if } s' = (1,2) \\ .1, & \text{if } s' = (1,1) \\ .1, & \text{if } s' = (2,1) \end{cases}$$

$$\delta((1,1), LEFT) = \delta(s' \mid (1,1), L) = \begin{cases} .9, & \text{if } s' = (1,1) \\ .1, & \text{if } s' = (2,1) \end{cases}$$

⋮

⋮

$(1,1) \Rightarrow R$

$(2,1) \Rightarrow R$

$(3,1) \Rightarrow U$

$(3,2) \Rightarrow U$

$$\pi: S \rightarrow A$$

policy

MARKOV PROPERTY: the next state S' is independent of the history, given the current state S (and the action a)

MDP

γ_0

P_0

R_0

V function

stole-value function

$\pi(s)$	G_1, P_1, R_1 $\vdots$ G_K, P_K, R_K $\vdots$	$V^\pi(s)$ $\underbrace{U^\pi(s)} = E \left[\underbrace{\sum_{t=0}^{\infty} R(S_t, \pi(S_t), S_{t+1})}_{\text{stole-value function}} \right]$ with $S_0 = s$ $\underline{S_0} = s \xrightarrow{\pi(s)} \underline{S_1}$
----------	--	---

$$V^{\pi_1}(\text{start}) = +\infty$$

$$V^{\pi_2}(\text{start}) = \dots$$

$$V(s|\sigma^*) = +\infty$$

DISCOUNT FACTOR $\gamma \in (0, 1]$

$$V\left(s_0 \xrightarrow{a_1} s_1 \xrightarrow{a_2} s_2 \rightarrow \dots \xrightarrow{a_n} \underline{s_n}\right) = \sum_{i=0}^{n-1} \gamma^i R(s_i, a_{i+1}, s_{i+1})$$

$$\underline{V(s_0 \xrightarrow{a_1} s_1 \xrightarrow{a_2} \dots)} = \sum_{i=0}^{\infty} \gamma^i R(s_i, a_{i+1}, s_{i+1}) \leq \frac{R_{\max}}{1-\gamma}$$

$\leq R_{\max}$

$\frac{R_{\max}}{1-\gamma}$

$$U: S \rightarrow \mathbb{R}$$

Γ_{∞}

Γ

$$V^\pi(s) \doteq E \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(s_t), s_{t+1}) \right]$$

↑
(STATE-) VALUE FUNCTION
V-function

$$\pi^* \text{ s.t. } V^{\pi^*}(s) \geq V^\pi(s), \forall s, \forall \pi$$

$$\pi^*(s) \doteq \underset{\pi}{\operatorname{argmax}} V^\pi(s) = \underset{\pi}{\operatorname{argmax}} E \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(s_t), s_{t+1}) \right]$$

OPTIMAL POLICY: π^* (is not unique)

$R(s, a, s')$

$V(s)$

3	0.576	0.078	0.953	+1
---	-------	-------	-------	----

11/11

$$U^{\pi^*}(s) = U(s)$$



Figure 17.3 The utilities of the states in the 4×3 world with $\gamma = 1$ and $r = -0.04$ for transitions to nonterminal states.

$$0.8 (0.8016) + 0.1 (0.6953) + 0.1 (7453) \quad \neq UP$$

$$0.1 (0.8016) + 0.8 (0.6953) + 0.1 (7453)$$

$$\pi^*(\dot{s}) = \arg \max_{a \in A} \sum_{s' \in S} p(s' | \dot{s}, a) \cdot \left(R(s, a, s') + \gamma \underbrace{U(s')} \right)$$

$\uparrow$
 UP, RIGHT

$$\pi^*(s) = \underset{a \in A}{\operatorname{argmax}} \sum_{s' \in S} p(s' | s, a) \cdot (R(s, a, s') + \gamma U(s'))$$

BELLMAN EQUATION

$$U(s) = \underset{a \in A}{\operatorname{max}} \sum_{s' \in S} p(s' | s, a) \cdot (R(s, a, s') + \gamma U(s'))$$

$$s \in S, |S| = n$$

$$U(s_0) = ? \quad U(s_1) = ? \quad \dots \quad U(s_n) = ?$$

- $U_0(s) = 0 \quad \underline{\forall s \in S}$

- $U_{t+1}(s) \leftarrow \max_{a \in A} \sum_{s' \in S} p(s' | s, a) \cdot (R(s, a, s') + \gamma U_t(s'))$

function VALUE-ITERATION(*mdp*, ϵ) **returns** a utility function

inputs: *mdp*, an MDP with states S , actions $A(s)$, transition model $P(s' | s, a)$,
rewards $R(s, a, s')$, discount γ

ϵ , the maximum error allowed in the utility of any state

local variables: U, U' , vectors of utilities for states in S , initially zero

δ , the maximum relative change in the utility of any state

repeat

$U \leftarrow U'; \delta \leftarrow 0$

for each state s **in** S **do**

$U'[s] \leftarrow \max_{a \in A(s)} \text{Q-VALUE}(\text{mdp}, s, a, U)$

if $|U'[s] - U[s]| > \delta$ **then** $\delta \leftarrow |U'[s] - U[s]|$

until $\delta \leq \epsilon(1 - \gamma)/\gamma$

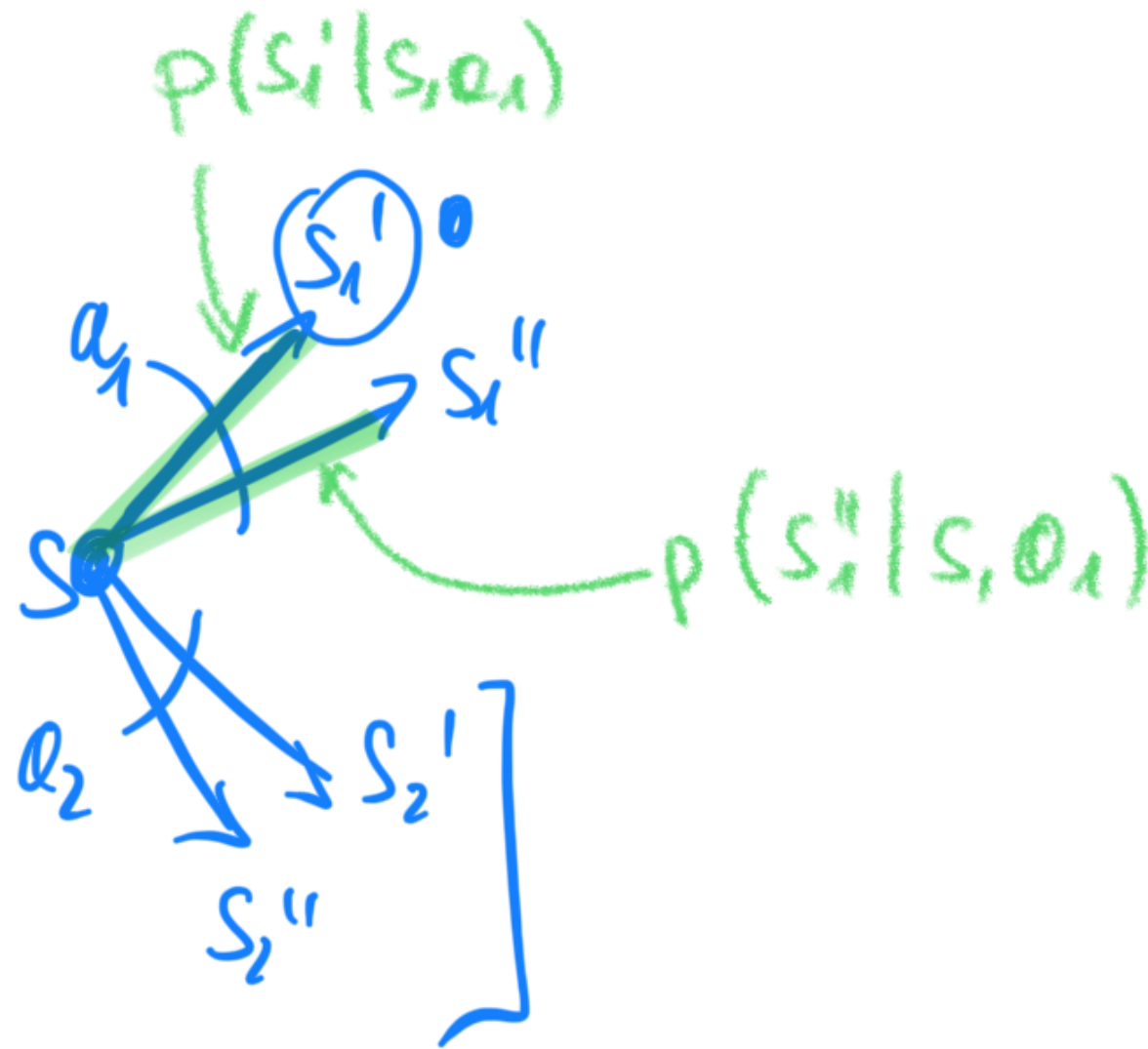
$$\sum_{s' \in S} p(s' | s, a) (R(s, a, s') + \gamma U(s'))$$

return U

Figure 17.6 The value iteration algorithm for calculating utilities of states. The termination condition is from Equation (17.12).

$$U_{t+1}(s) \leftarrow \max_{a \in A} \sum_{s' \in S} p(\underline{s'} | s, a) \cdot \left(\underline{R(s, a, s')} + \gamma \overbrace{U_t(s')}^0 \right)$$

$$\left\{ \begin{array}{l} a = R \quad (.8 \cdot 1) + .1(0) + .1(0) = (.8) \\ a = D \quad (.8 \cdot 0) + .1 \cdot .8 + .1(0) = .08 \\ \vdots \end{array} \right.$$



Expected utility when performing a_1 in s

$$p(s_1'|s, a_1) [R(s, a_1, s_1') + \gamma U(s_1')] +$$

$$p(s_{n+1} | s_n, a_n) [R(s_n, a_n, s_{n+1}) + \gamma U(s_{n+1})]$$

• POLICY ITERATION

(Sutton, Barto,
Reinforcement Learning
an introduction, 2020, p. 80)

→
- POLICY EVALUATION: given a policy π , computes
the corresponding V-function U^π

- POLICY IMPROVEMENT: based on U^π , improve π

1. Initialization:

$V^\pi(s) \in \mathbb{R}$ and $\pi(s) \in A$ arbitrarily, for all $s \in S$;

$$V^\pi(\text{terminal}) = 0;$$

2. Policy Evaluation:

Loop:

$$\Delta \leftarrow 0;$$

For all $s \in S$:

$$v \leftarrow V^\pi(s); \quad // v: \text{value before update}$$

$$V^\pi(s) \leftarrow \sum_{s'} p(s'|s, \pi(s)) \left(R(s, \pi(s), s') + \gamma V^\pi(s') \right)$$

$$\Delta \leftarrow \max(\Delta, |v - V^\pi(s)|)$$

until $\Delta < \epsilon$

3. POLICY IMPROVEMENT
policy-stable $\leftarrow$ true;

For each $s \in S$:

old-action $\leftarrow \pi(s)$

$$\pi(s) \leftarrow \underset{a \in A}{\operatorname{argmax}} \sum_{s' \in S} p(s' | s, a) (R(s, a, s') + \gamma V^\pi(s'))$$

if (old-action $\neq \pi(s)$) then policy-stable $\leftarrow$ false;

If policy-stable return $V^\pi \approx V$ and $\pi \approx \pi^*$

else repeat from 2.

PLANNING

Full observability

- DETERMINISTIC $\circ$
- NON DETERMINISTIC $\circ$
- STOCHASTIC $\circ$



$$\pi(s) \leftarrow a$$



- Q-function : $Q^\pi(s, a)$: function returning expected utility for executing a in S and then following π

$$\sum_{s' \in S} p(s' | s, a) (R(s, a, s') + \gamma V^\pi(s'))$$



State-action value function of π

$$\underline{Q^{\pi}(a, s)}$$

Q-function

$$Q^{\pi^*}(s, a) = Q(s, a)$$

Q-function of
optimal policy π^*

$$Q(s, a)$$

$$\pi^*(s) = \underset{a}{\operatorname{argmax}} Q(s, a)$$
